The Vertex-Minimal Nontrivial Nucleus: An Exhaustive Computational Classification

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Supplementary computational details for *Counting Triangulated 3-Spheres, Nuclei, and Polyhedra*.

1 Nuclei and the search bounds

A *nucleus* is a finite simplicial triangulation N of the 3-ball such that

(i) every vertex of N lies on ∂N, and

(ii) every interior triangle contains at most one boundary edge.

It is *nontrivial* if it has more than one tetrahedron. Write $v = f_0(N)$ and $t = f_3(N)$. An edge is *interior* if it does not belong to ∂N; since every vertex belongs to the boundary, interior edges are also called spanning edges.

The classification below uses one geometric input, proved independently in the structural part of the companion paper:

> *All-interior-edge tetrahedron theorem.* Every nontrivial nucleus contains a tetrahedron all six of whose edges are interior.

This result is not inferred from the enumeration. It provides a necessary constraint for the search; none of the arguments below requires the converse.

Lemma 1. *For a triangulated 3-ball N whose vertices all lie on its boundary,*

$$f_1^{\mathrm{int}}(N) = t - v + 3.$$

Proof. The boundary is a triangulated 2-sphere on v vertices, so

$$f_1(\partial N) = 3v - 6, \qquad f_2(\partial N) = 2v - 4.$$

Counting tetrahedron–triangle incidences gives

$$4t = 2f_2^{\mathrm{int}}(N) + 2v - 4, \qquad f_2(N) = 2t + v - 2.$$

Euler's formula $v - f_1(N) + f_2(N) - t = 1$ now yields $f_1(N) = t + 2v - 3$. Subtracting the $3v - 6$ boundary edges proves the claim. □

Proposition 2. *If N is a triangulated 3-ball with all vertices on its boundary, then*

$$\boxed{t \le \binom{v}{2} - 2v + 3.}$$

In particular, $t \le 28$ for $v \le 10$, and $t \le 36$ for $v = 11$.

Proof. At most $\binom{v}{2}$ edges occur in N, of which $3v-6$ lie on the boundary. Lemma 1 thus gives

$$t - v + 3 = f_1^{\mathrm{int}}(N) \le \binom{v}{2} - (3v-6),$$

which rearranges to the stated bound. Its right-hand side is increasing for $v \ge 3$ and takes the values 28 and 36 at $v = 10$ and $v = 11$. □

For $v = 11$, the all-interior-edge tetrahedron theorem also gives $f_1^{\mathrm{int}}(N) \ge 6$. Lemma 1 then implies

$$14 \le t \le 36. \tag{1}$$

2 Enumeration of the possible boundaries

The boundary of a nucleus is a simplicial triangulation of the 2-sphere on the same vertex set. Its combinatorial types are generated using `plantri`, developed by Gunnar Brinkmann and Brendan McKay. A maximal planar graph with its planar embedding determines a simplicial triangulated 2-sphere. For at least four vertices the embedding is unique up to reflection, so the triangulation mode of `plantri` lists the possible boundary spheres once up to isomorphism.

The numbers of boundary types are

v	4	5	6	7	8	9	10	11
$\#S$	1	1	2	5	14	50	233	1249.

Thus there are 306 types for $4 \le v \le 10$ and 1249 types for $v = 11$. The lists used in this classification are included in the reproducibility package as `data/boundaries_v4.asc` through `data/boundaries_v11.asc`. The `plantri` program can be obtained separately from its official distribution at `https://users.cecs.anu.edu.au/~bdm/plantri/`. Different software versions may list the same combinatorial types in a different order; the original input files are therefore retained.

3 A necessary binary filling model

Fix a boundary sphere S on vertex set V, and write $E(S)$ for its edges. Any triangle $F \notin S$ containing two or three edges of $E(S)$ is *forbidden*: if it belonged to a selected tetrahedron, the prescribed boundary incidences would force it to be an interior triangle, contrary to the nucleus condition. We therefore discard every candidate tetrahedron containing a forbidden triangle.

For each remaining four-element set $Q \subseteq V$ introduce a binary variable x_Q, recording whether Q is selected as a tetrahedron. For each nonboundary, nonforbidden triangle F introduce a binary variable y_F, recording whether it is an interior triangle. For each nonboundary edge e introduce a binary variable z_e, recording whether it occurs in the filling. The model imposes the following necessary conditions.

(1) **Boundary triangles.** Every triangle of S is incident with exactly one selected tetrahedron:

$$\sum_{Q \supset F} x_Q = 1 \qquad (F \in S).$$

(2) **Nonboundary triangles.** Every allowed triangle outside S is incident with zero or two selected tetrahedra:

$$\sum_{Q \supset F} x_Q = 2y_F \qquad (F \notin S \text{ allowed}).$$

Forbidden triangles occur in no candidate tetrahedron.

(3) **Nonboundary edges.** For every $e \notin E(S)$, with Q ranging over candidate tetrahedra containing e, impose

$$x_Q \le z_e \quad (e \subset Q), \qquad z_e \le \sum_{Q \supset e} x_Q.$$

Consequently $z_e = 1$ if and only if e occurs in a selected tetrahedron.

(4) **Tetrahedron bound.** Proposition 2 yields

$$\sum_Q x_Q \le \binom{v}{2} - 2v + 3.$$

(5) **An all-interior-edge tetrahedron.** The independent structural theorem gives

$$\sum_{Q:\, \binom{Q}{2} \cap E(S) = \varnothing} x_Q \ge 1.$$

If no candidate Q has all six edges outside $E(S)$, the system is immediately infeasible.

(6) **Vertex-link Euler characteristics.** Every vertex link in a triangulated 3-ball with all vertices on the boundary is a disk and has Euler characteristic one. The vertices, edges, and triangles of $\mathrm{lk}_N(a)$ correspond to the edges, triangles, and tetrahedra of N containing a. Thus, for each $a \in V$,

$$\#\{e \in E(S) : a \in e\} + \sum_{\substack{e \notin E(S) \\ a \in e}} z_e - \#\{F \in S : a \in F\} - \sum_{\substack{F \notin S \text{ allowed} \\ a \in F}} y_F + \sum_{Q \ni a} x_Q = 1.$$

These conditions are *necessary*, not asserted to be sufficient. Every nontrivial nucleus with boundary S determines a feasible binary assignment by taking the indicators of its tetrahedra, interior triangles, and interior edges. Conversely, a feasible assignment may fail to triangulate a 3-ball. Notice also that once the tetrahedron indicators x_Q are fixed, the triangle and edge equations uniquely determine all y_F and z_e. This will justify blocking solutions using only the x_Q variables.

4 The lower bound of eleven vertices

For each of the 306 boundary spheres with $4 \le v \le 10$, the model in Section 3 was solved as a binary feasibility problem using SciPy's interface to HiGHS. All 306 instances were reported infeasible. The boundary files and machine-readable feasibility results are included in the reproducibility package.

Theorem 3. *Every nontrivial nucleus has at least* 11 *vertices.*

Proof. Suppose a nontrivial nucleus N has $v \le 10$ vertices, and put $S = \partial N$. Up to relabeling, S is one of the 306 boundary spheres from Section 2. The tetrahedra, interior triangles, and interior edges of N determine a feasible assignment for the corresponding binary model, by the observations in Section 3. But the exhaustive computation reports all such models infeasible, a contradiction. □

The argument does not need 3-ball recognition: infeasibility already holds for a system of necessary combinatorial constraints.

5 Classification on eleven vertices

There are 1249 possible boundary types, and by (1) it suffices to consider $14 \le t \le 36$. For each boundary S and each t in that range, the model is augmented by

$$\sum_{Q} x_Q = t. \tag{2}$$

Whenever the solver finds a feasible tetrahedron indicator vector $\bar{x} \in \{0,1\}^m$, the program appends the *no-good* inequality

$$\sum_{i:\bar{x}_i=1} x_i - \sum_{i:\bar{x}_i=0} x_i \le |\{i : \bar{x}_i = 1\}| - 1.$$

This excludes exactly that tetrahedron set. As noted above, the y and z variables are uniquely determined by x, so no different filling is lost. Iteration continues until HiGHS reports infeasibility; hence the procedure enumerates all feasible binary fillings, subject to the solver's reported decisions.

For each enumerated filling, the program checks the exact simplicial incidences and local links and computes homology over $\mathbb{F}_2$. Every 3-ball has Betti numbers $(1,0,0,0)$, so any other Betti vector excludes the filling. The reported exhaustive counts are:

t	feasible binary fillings	nuclei
14–29	0	0
30	8	0
31	54	0
32	213	0
33	504	0
34	623	0
35	360	1
36	67	0
Total	1829	1

Of the 1829 feasible fillings, 1828 have mod-2 Betti numbers $(1,1,1,0)$ and therefore are not 3-balls. The only remaining filling occurs at $t = 35$, on the stored boundary representative numbered 689 (using the program's zero-based indexing). Its 35 facets are recorded in `results/n00_facets.json`. We call it $N_{\min}$; it is the complex denoted `N00` in the computation. Its vertices are labeled $0, \dots, 10$ in the data, corresponding to $1, \dots, 11$ in the main paper. A direct computation gives

$$f(N_{\min}) = (11, 54, 79, 35), \qquad f(\partial N_{\min}) = (11, 27, 18).$$

The positive verification is separate from the homology rejection of the other fillings. The facet list passes exact checks of the prescribed spherical boundary, triangle incidences, reducedness, dual connectivity, vertex-disk links, and the appropriate interval or circle edge links. In addition, a certificate of 89 elementary simplicial collapses to a vertex is supplied and independently replayed. The stricter replay verifier also checks that a supposedly free face has no additional cofaces in higher dimensions. These checks show that the complex is a contractible PL 3-manifold with boundary S^2 and hence a PL 3-ball. Thus it is indeed a nucleus.

Theorem 4 (Classification at the minimum)**.** *Up to simplicial isomorphism, $N_{\min}$ is the unique nontrivial nucleus with* 11 *vertices. It has exactly* 35 *tetrahedra and*

$$f(N_{\min}) = (11, 54, 79, 35).$$

In particular, there is no nontrivial 11*-vertex nucleus with* 36 *tetrahedra.*

Proof. Let N be any nontrivial 11-vertex nucleus. Its boundary agrees, after relabeling, with one of the 1249 generated spheres, and its number of tetrahedra satisfies $14 \leq t \leq 36$. Therefore its tetrahedra induce one of the feasible binary fillings exhaustively enumerated for that boundary and t. Every filling other than the displayed $N_{\min}$ has nonzero first and second homology over $\mathbb{F}_2$, whereas $N_{\min}$ is positively verified to be a nucleus. This proves uniqueness and the stated facet count. □

Corollary 5. *The minimum number of vertices of a nontrivial nucleus is* 11*, and there is exactly one combinatorial type at this minimum:* $N_{\min}$*.*

Proof. Theorem 3 rules out fewer vertices, while Theorem 4 establishes existence and uniqueness at 11. □

6 Reproducibility and scope of the certificates

The reproducibility package contains the stored `plantri` outputs, not the program's source, which is available separately from its official website. It contains the common binary-model implementation, the $v = 11$ classification driver, the 1249 stored boundary representatives, the recorded $v = 11$ count table, the facet list and collapse certificate for $N_{\min}$, and their checksums. It also contains the boundary data and complete machine-readable results for the $v \leq 10$ lower-bound run, together with a verifier of the collapse certificate.

From the package root, the principal commands are:

```
# Obtain and compile plantri from its official website first.
PLANTRI=/absolute/path/to/plantri
python src/nucleus_lower_bound.py \
    --plantri "$PLANTRI" \
    --min-v 4 --max-v 10 --out reproduced_lower_bound
python src/classify_v11.py \
    --plantri "$PLANTRI" \
    --out reproduced_v11.json
python src/verify_n00_collapse_strict.py \
    results/n00_collapse_certificate.json
sha256sum -c SHA256SUMS.txt
```

Remark 6 (Computational status)**.** *The $v \leq 10$ infeasibility results and the positive $N_{\min}$ checks are reproducible by the supplied programs. The $v = 11$ counts are provided as machine-readable summaries of the exhaustive enumeration; they can be recomputed with the included driver. The negative results do not include independently checkable solver proof objects.*

The negative conclusions depend on the correctness of `plantri`*, the binary-model encoding, and the HiGHS feasibility and infeasibility decisions. The package does not contain independently checkable certificates of infeasibility for each MILP instance. In contrast, the positive collapse certificate is replayable independently of the MILP solver. These qualifications apply to the computational classification theorem above.*